

Bayes classifier: Exercise

- Educational experts have built a model to predict whether an incoming freshman will pass or fail their introductory calculus class, based on the number of hours/day spent studying the lectures (S) and doing the homework (H). Using a binary random variable Y to code for pass/fail, the model is given by:

$$\bullet \quad Y = \begin{cases} 1 \text{ (pass)} & \text{if } S + H + N > 5, \\ 0 \text{ (fail)} & \text{otherwise} \end{cases}$$

- for $S, H, N \geq 0$, where N is an unobservable variable corresponding to factors such as motivation, here translated to the equivalent in hours/day. The variables S , H , and N are modelled as independent and exponential with parameter $\lambda = 1$. Find the optimal predictor of whether a given student will pass the class based on S and H .
- *Hint: Find the posterior probability function $\eta(\mathbf{x}) = P(Y = 1 \mid \mathbf{X} = \mathbf{x})$ and apply*

$$\bullet \quad \psi^*(\mathbf{x}) = \begin{cases} 1, & \eta(\mathbf{x}) > 0.5 \\ 0, & \text{otherwise} \end{cases}$$